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EDUCATION

Aarhus University

Master of Science in Computer Science

Aarhus, Denmark

Aug. 2024 – Jun. 2026

- Focused on cryptography, machine learning, artificial intelligence, and trustworthy AI, combining theoretical foundations with implementation-driven coursework.
- Completed a master's thesis on spacing-based watermarking for large language model output, exploring keyed linguistic events, error-correcting codes, synchronization, and robustness to text modifications.

Poznań University of Economics and Business

Bachelor of Science in Computer Science and Econometrics – Electronic Business

Poznań, Poland

Oct. 2020 – Jul. 2024

- Combined programming, databases, statistics, econometrics, and data analysis to solve practical business and economic problems.
- Graduated with a specialization in Electronic Business, focused on applying IT systems, data, and digital technologies to business processes and decision-making.

EXPERIENCE

Pharmacy Operations Assistant

Apteka Elias sp. z o.o. – Seasonal

May 2022 – Sep. 2025

Ustronie Morskie, Poland

- Coordinated supplier orders, incoming deliveries, invoices, and operational documentation while following established pharmacy procedures.
- Monitored stock levels and product availability, identified shortages, and supported timely replenishment during high-demand summer periods.
- Organized daily workflows and supported seasonal staff to keep customer service and back-office operations running efficiently.
- Translated knowledge of pharmacy scheduling requirements into the Work Scheduler application, which is currently used in the daily operations of one pharmacy.

Maritime Lifeguard

Ustronie Morskie Commune – Seasonal

Jun. 2018 – Aug. 2021

Ustronie Morskie, Poland

- Maintained beach safety through continuous risk assessment, preventive communication, and rapid response to changing sea and weather conditions.
- Conducted water rescue operations, including interventions as a jet-ski operator, and provided first aid when immediate medical assistance was required.
- Worked within a coordinated rescue team, communicating clearly and making time-critical decisions under pressure.

PROJECTS

[Spacing-Based Watermark](#) | *Python, PyTorch, Hugging Face*

Ongoing

- Developing a research implementation for embedding and detecting spacing-based watermarks in text produced by large language models.
- Designed a pipeline using keyed function-word events, modulo spacing residues, LDPC error correction, and marker-based synchronization.
- Built experiments to evaluate watermark recovery after insertions, deletions, substitutions, and other text modifications.

[Work Scheduler](#) | *Python, PySide6, SQLite*

Ongoing

- Built a local-first, cross-platform desktop application for creating and managing pharmacy staff schedules.
- Implemented Polish staffing rules, public-holiday handling, bilingual interfaces, database migrations, and exportable PDF reports.
- Added automated tests and multi-platform release workflows; the application currently supports daily operations in one pharmacy.

[Surfing Project](#) | *Django, PostgreSQL, Docker*

Ongoing

- Developing a full-stack social platform for discovering surf spots, organizing sessions, and connecting with other surfers.
- Implemented user profiles, follows, session participation, photo sharing, direct and group chat, notifications, and role-based permissions.

- Structured the application around a relational PostgreSQL data model and a containerized development environment, supported by automated tests.

US Accidents Visualization Dashboard | *Python, Dash, Plotly, Polars*

Completed

- Created an interactive dashboard for exploring United States road accidents from 2016–2023 across location, severity, time, road, and weather dimensions.
- Built linked charts and maps that allow users to filter the dataset and compare geographic and temporal patterns.
- Used Polars, Pandas, and cached processing to prepare a large real-world dataset for responsive exploration.

CERTIFICATIONS

Python Developer Programme – Coders Lab

Certificate 334/12/2022

Python, SQL, PostgreSQL, Django, Flask, HTML/CSS, JavaScript fundamentals, Git, Scrum

2022

- Completed intensive, project-based training in Python development, relational databases, backend frameworks, web fundamentals, testing, and version control.
- Built web applications using Django and Flask, with PostgreSQL, HTML/CSS, and the fundamentals of JavaScript for client-side interactions.
- Worked through iterative development modules using Git and Scrum-style planning and delivery practices.

C1 Advanced (CAE) – Cambridge English

Certificate C7185846

English language proficiency – CEFR level C1

2024

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript fundamentals, HTML/CSS

Backend and Applications: Django, Flask, REST API, Fast API, PySide6, Dash

Machine Learning and AI: PyTorch, TensorFlow, scikit-learn, Hugging Face, large language models

Data and Visualization: Pandas, Polars, NumPy, Plotly, data analysis, data visualization

Databases and ORM: PostgreSQL, SQLite, SQLAlchemy, Alembic

Tools and Delivery: Docker, Git, GitHub, GitHub Actions, pytest, CI/CD

Ways of Working: Agile development, Scrum, iterative delivery, collaborative version control

LANGUAGES

Polish: Native | **English:** C1 | **Danish:** A2